

CHROMATE

_____Name

_____Date

*also includes chromium and its compounds, like **potassium dichromate**, ammonium dichromate, chromated zinc chloride, chromic acid, chromic nitrate, chromium dioxide, chromium hydroxide ultramarines, chromium phosphate, chromium oxide, chromium potassium sulfate, potassium bichromate, sodium dichromate, and zinc chromate.*

What is it?

Chromium is an element that is present in small amounts in many substances. Its many uses include for green color, to tan leather, to harden cement, and to make metal alloys.

Where might it be found?

Cosmetic pigments (mascara, eye shadow, eye liner, concealer, foundation, powder)
Face scrub or mask
Soap bars (e.g. Irish Spring)
Shaving cream (prevents rust)

Leather, tanned (glove, golf glove, boots, shoes, garments, accessories, upholstery, briefcase, purse, wallet, belt, suede item, steering wheel)
Metallurgy, electroplating, galvanizing, welding
Antioxidant for antifreeze, radiators, pipelines
Anticorrosive in dips, oils, fuels, drilling muds
Explosives, fireworks, safety match heads
Animal hide glues
Liner in high temperature industrial furnaces

Also in...

Ceramic & artist pigments
Tattoos (especially green)
Textile dye fixing agent
Roulette & pool table green felt
Shoe polish
Floor wax
Chromic catgut and silk suture
Orthopedic prostheses
Acupuncture needles
Orthodontic wires
Dental fillings, implants
Chrome-nickel dental crowns
Detergents, esp. phosphate
Photographic chemicals
Photocopy paper, blueprints
Engraving, lithography, offset
Magnetic tapes
Wet cement, mortar, bricks and
Epoxy filler, spackle
Drywall, primer, paint
Pressure treated wood
Fungicide in grout
Veterinary fungicide
Milk testing, lab test substance
Gun & carburetor degreaser

How to avoid it:

Avoid using any product that contains chromates. It is important to check the complete ingredient list of each product you use for a name listed above. Some products' ingredient lists are available on the internet at the manufacturer's website, the store's website, or websites such as <http://householdproducts.nlm.nih.gov>, <http://www.cosmeticsdatabase.com>, or <http://www.drugstore.com>.

For possible workplace exposure, check the Material Safety Data Sheet (MSDS) and the ingredient labels of all solutions and products encountered at work.

Replace leather with vinyl, polyvinylchloride (e.g. Tingley rubber work boot), canvas (e.g. Ked), or synthetics. Or use aldehyde- or vegetable-tanned leather (e.g. www.mooshoes.com or vegetarian choices at www.zappos.com). Some Hush Puppies (800-545-2425), Birkenstock, and Ecco are chromate-free. Instead of leather gloves, wear heavy duty fabric, canvas, rubber, or vinyl. For dental crowns, use porcelain. For orthopedic prostheses, consider titanium. *(continued)*

NOTE: Problems at work can come from contact with the chromium fumes in processes like welding and electroplating. Seek protection by using an exhaust hood, respirator mask, or other means of proper ventilation or shielding.

Chromate dermatitis is known for being thick and longstanding. Rash may persist long after stopping contact with the metal.

Sometimes it is necessary to avoid **oral** sources of chromium. Your doctor may advise you to try for one month a diet that eliminates these items. If your dermatitis improves, you may try adding back one item per week, watching for worsening.

DIETARY SOURCES OF CHROMIUM TO AVOID:

Multivitamins (*check the ingredients on the label*)
Chromium picolinate
Apple peel
Beans, green or baked
Beer (brewer's yeast)
Black pepper
Broccoli
Canned fruits (plums)
Cereals high in bran
Chicken, eggs
Chocolate
Cloves and other spices
Coffee
Corn
Frozen or canned vegetables
Mushrooms
Onions
Potatoes
Processed meats (beef)
Seafood (oysters, mussels, clams, cockles)
Tea
Thyme
Watercress
Wine